REPUBLIC POLYTECHNIC

Lesson 13
Simulation Experimentation

E211 – Operations Planning II

Bottling Line 3D Model

Help Justin study these additional proposed changes:

A. Install a conveyor to move items between Pack and Palletize. A Robot arm will perform the palletization. Pack will be fully automated (process time remains unchanged) so that there no need for setup. Operator2 is redeployed to perform unpacking at a newly setup station in parallel to the current unpack station.

B. To support reusables, the company wants to increase the use of recycled bottles. It is estimated that the QC pass rate will drop to 80% and more thorough cleaning for bottles is required. CleanSterlizeDry process time will follow Uniform Distribution (min = 20s, max = 30s).

C. To increase recycling rate, the maximum number of reworks allowed is increased to 2. For bottles that have failed QC 2 times, CleanSterlizeDry process time will be increased to a constant time of 45s (timing for 1st and 2nd cleaning follows the timing given previously). Assign the color black to bottles that fail 3 times (reclaimed bottles).

Food Ordering Process Model

A. Determine the most efficient set up of staff and their role assignment to achieve these performance measures:

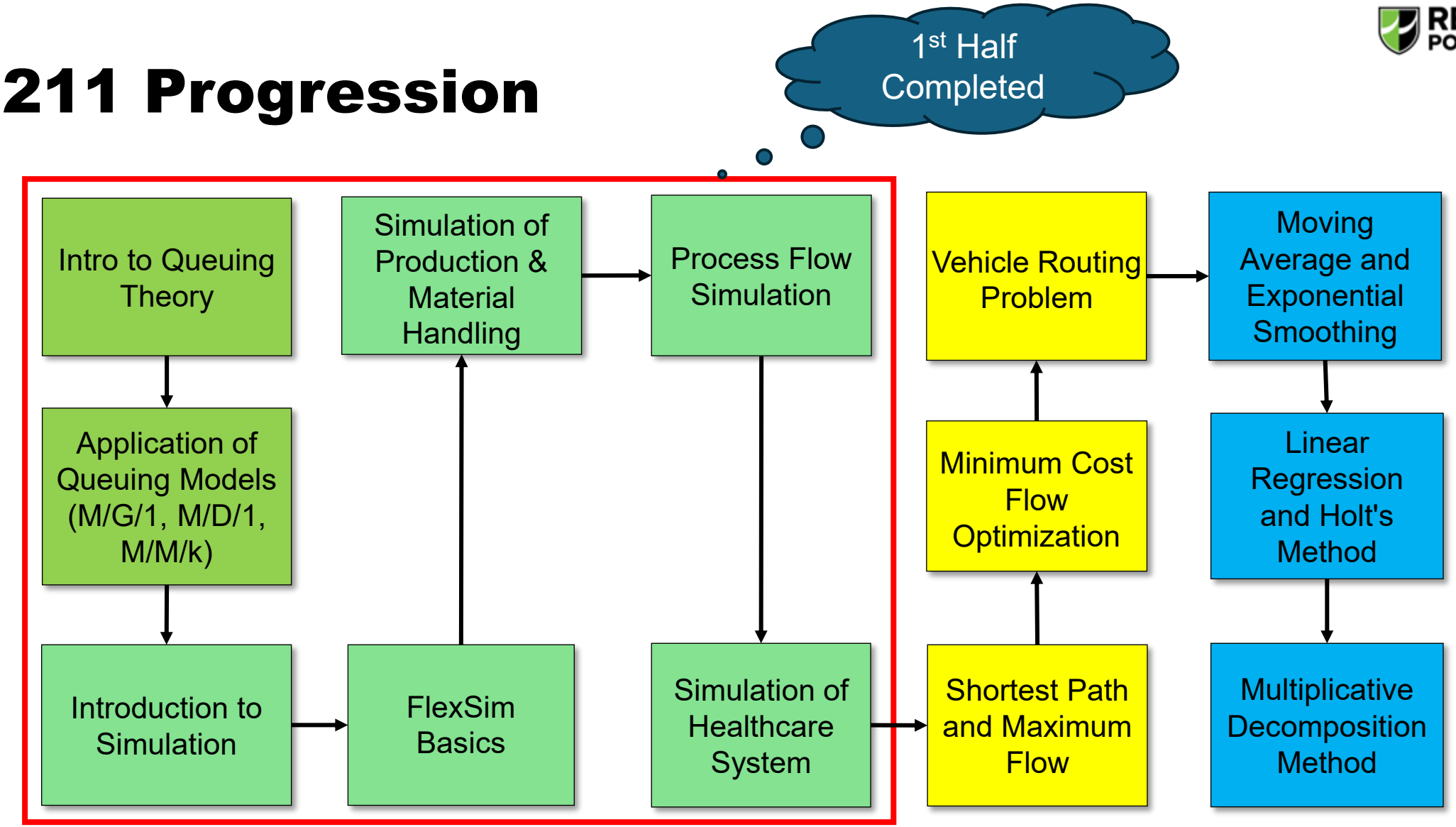
Mean Time in System ≤ 10mins
Mean Number of Customers served ≥ 45

B. Determine the total space required to accommodate customers in the system for your solution to part (A).

REPUBLIC POLYTECHNIC

Clinic Healthcare Model

- Following patients' complaints about the long stay in the clinic, the clinic decides to set a service standard where the mean Time in System for patients should be 60 mins or less.

- Determine what changes are required to achieve the performance target set by the clinic. All patients should complete their treatment by 5pm.

E211 Progression